

W1 (the genuine open core: construct 4D YM as a rigorous QFT) — attack-framing via the HS route, per Casey’s flag. HONEST: this SCOPES the wall, it does NOT solve constructive 4D QFT (open in mathematics itself). The contribution: the HS isometry relocates the constructive burden off flat- R^4 (the famous open problem) onto the BSD bulk (rigorous harmonic analysis) + exact HS transfer, encoding the interaction in the bulk curvature; and it names the residual precisely — ‘rigorous BSD-holographic map + YM identification’ = the BST-specific version of making AdS/CFT rigorous. Sharp caveat flagged: the relocation risks delivering rigor for a GENERALIZED-FREE boundary theory, not interacting YM — that risk IS the residual.

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W1 — the constructive core, framed via the HS route

W1 = construct 4D Yang-Mills as a rigorous (Wightman/Osterwalder-Schrader) QFT and prove the gap. This is open in mathematics itself — no interacting 4D QFT has ever been constructed. The HS route does not solve it. What it does is *relocate* the constructive burden to where BST has exact tools, and name the residual precisely. This note is the attack framing; W1 stays open.

The three ingredients, by where their rigor stands

1. The bulk D_{IV}^5 spectral theory = harmonic analysis on a bounded symmetric domain. Rigorous and classical (Hua-Korányi, Faraud-Korányi): the Hardy space H^2 , the Casimir spectrum, the Szegő kernel — all textbook, not an open construction. The gap = C_2 is an exact eigenvalue.
2. HS (T2489) = exact unitary interior $\leftrightarrow$ Shilov boundary. Transfers the bulk data exactly: Hilbert space, Hamiltonian (Casimir), spectrum, and — the L3 lead — Szegő-kernel positivity = OS reflection positivity. So the boundary inherits a rigorous positive-energy, reflection-positive structure from the bulk, for free.
3. The boundary object = 4D conformal YM (UV fixed point). The identification “this boundary theory IS YM” is the load-bearing step.

The relocation (the useful structural point)

- Direct W1: construct interacting YM on flat R^4 — the famous open-in-mathematics problem.

- HS-routed W1: (a) the bulk is rigorous BSD harmonic analysis [have it] + (b) HS transfers it exactly [have it, T2489] + (c) identify the transferred boundary theory as YM.

So the constructive rigor lives on the bulk side (tractable), and the interaction is encoded in the bulk curvature (Casimir gap; Bergman curvature $-2/g$) — the standard holographic “classical bulk $\leftrightarrow$ interacting boundary.” This is why R^4 “can’t host the scale” (the no-go) but the bulk can: the bulk *is* where the scale and the dynamics live.

What W1 reduces to (honestly)

W1 $\rightarrow$ “is the holographic map (rigorous BSD bulk) $\rightarrow$ (boundary 4D QFT) itself rigorous, and is the image 4D YM?” — the BST-specific version of *making AdS/CFT rigorous*. Two halves:

- The identification-as-YM half = the W2 spectrum match run *to completeness* (the dynamics/correlators, not just the gap) — the same computation the team is closing.
- The rigorous-holographic-map half = the genuine residual. It is *more tractable here than generic AdS/CFT* because: the bulk is a BSD with exact harmonic analysis (not a generic AdS), HS is an exact isometry (not an asymptotic dictionary), and OS-positivity is already in hand (positive Szegő kernel).

🔍 The sharp caveat (the real edge — flagged, not buried)

The relocation risks delivering rigor for the WRONG theory. In AdS/CFT, a *free* bulk field gives a generalized-free boundary theory — rigorous, but *not* an interacting CFT. The D_IV⁵ “spectral theory” (harmonic analysis) is free-field-like (a discrete Casimir spectrum). So the live danger is: HS transfers the bulk exactly, but the boundary you get is generalized-free, not interacting Yang-Mills. For the image to be YM, the interaction must genuinely be encoded — in the bulk curvature / the non-trivial K-type structure / bulk vertices — and *proving the boundary correlators are interacting (not generalized-free)* is the heart of the residual. This is exactly where the W2 spectrum match to completeness becomes load-bearing for W1: a full spectrum with the right *interacting* glueball structure (anomalous dimensions, correct correlator ratios) is the evidence the boundary isn’t generalized-free. So W1’s hard half and W2’s completeness are the same question seen from the constructive side.

Honest tier

- This is the attack framing, NOT a proof. W1 stays OPEN.
- The contribution (legitimate, structural): it relocates the constructive burden off flat- R^4 (the famous open problem) onto the BSD bulk + exact HS transfer (where BST has unusual tools — exact harmonic analysis, exact isometry, in-hand OS positivity); it names the residual precisely; and it flags the genuine risk (generalized-free vs interacting), tying it to the W2 completeness check.
- It does NOT claim to construct YM. It claims the HS route is the right line of attack, says *why* (the tools), and says *where it can still fail* (the free-vs-interacting edge). Count UNAFFECTED 4 of 26.

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